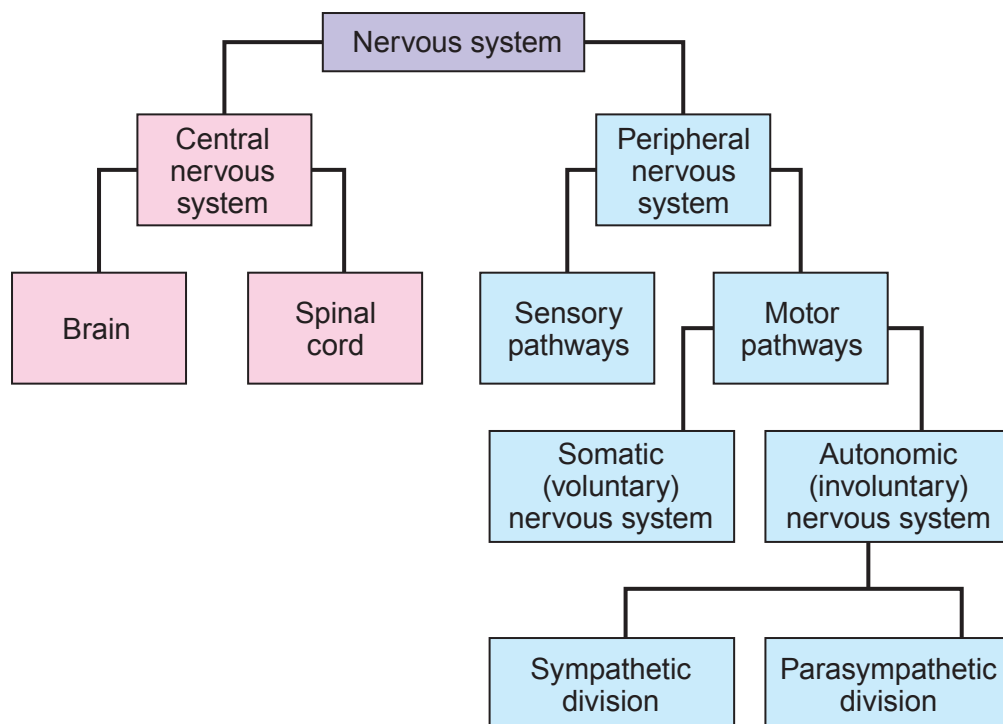


Option C: Neurobiology and Behaviour

10. (a) The diagram below shows the organisation of the human nervous system.

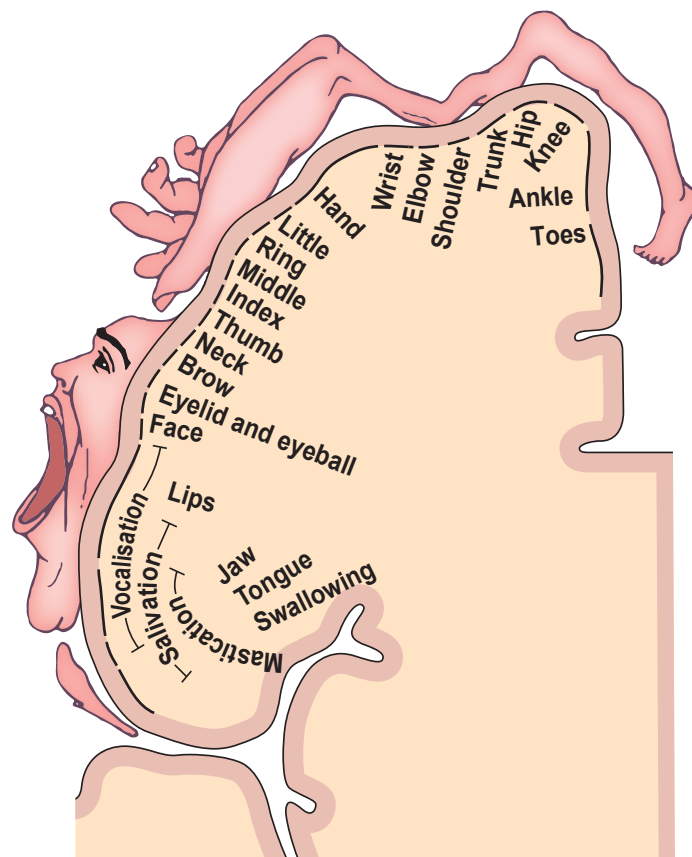


- (i) Complete the table below to show the differences between the divisions of the autonomic nervous system. [3]

	Sympathetic	Parasympathetic
Neurotransmitter		
General effect		
Effect on heart rate		



The somatic nervous system is involved in controlling voluntary movements via motor neurones, and sensory information via sensory neurones. This links to the motor and sensory cortex of the cerebrum of the brain. The diagram below shows a motor homunculus.



(ii) Explain what the motor homunculus represents.

[1]

.....

.....

(iii) Suggest why the sensory homunculus would differ from the motor homunculus.

[1]

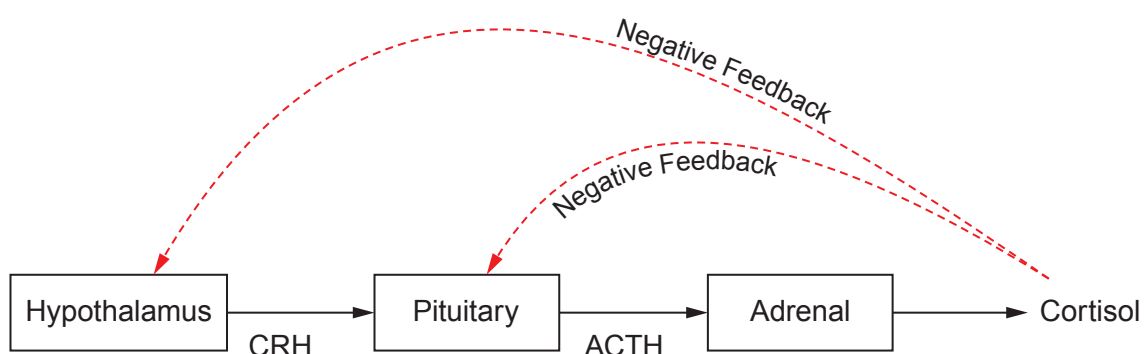
.....

.....



- (b) The hypothalamus is the link between the nervous system and the endocrine system, and is found in the brain. It is closely associated with the pituitary gland.

Cortisol is a hormone produced by the adrenal glands and is involved in the body's response to stress. The normal release of cortisol is controlled by the hypothalamus as shown in the diagram below.



High levels of cortisol could be a cause of mental illness. Cushing's disease can be caused by a tumour on the pituitary gland, which causes an abnormally high level of ACTH to be released.

- (i) Using the information above, and your own knowledge, suggest why mental illness has been linked with Cushing's disease. [3]

.....

.....

.....

.....

- (ii) The gene for cortisol is expressed in the cells of the adrenal gland. Suggest how the expression of this gene could be affected in children who have experienced trauma. [2]

.....

.....

.....

.....



- (c) Behaviour describes many actions by living organisms in relation to other organisms or the environment they inhabit. Behaviours can be highly complex or seem quite simple.

Ants are social insects which live in large colonies.

- (i) State the advantages to the ants of living in a colony. [2]

.....

.....

.....

- (ii) Suggest how ants may communicate with each other. [1]

.....

.....

Sexual selection has played a role in the evolution of courtship behaviours in vertebrates. Male elephant seals fight with other males for breeding rights, often with fatal consequences.

The photograph below shows two male elephant seals fighting.



- (iii) State the type of sexual selection displayed by the male elephant seals. Explain the consequence of this type of selection. [2]

.....

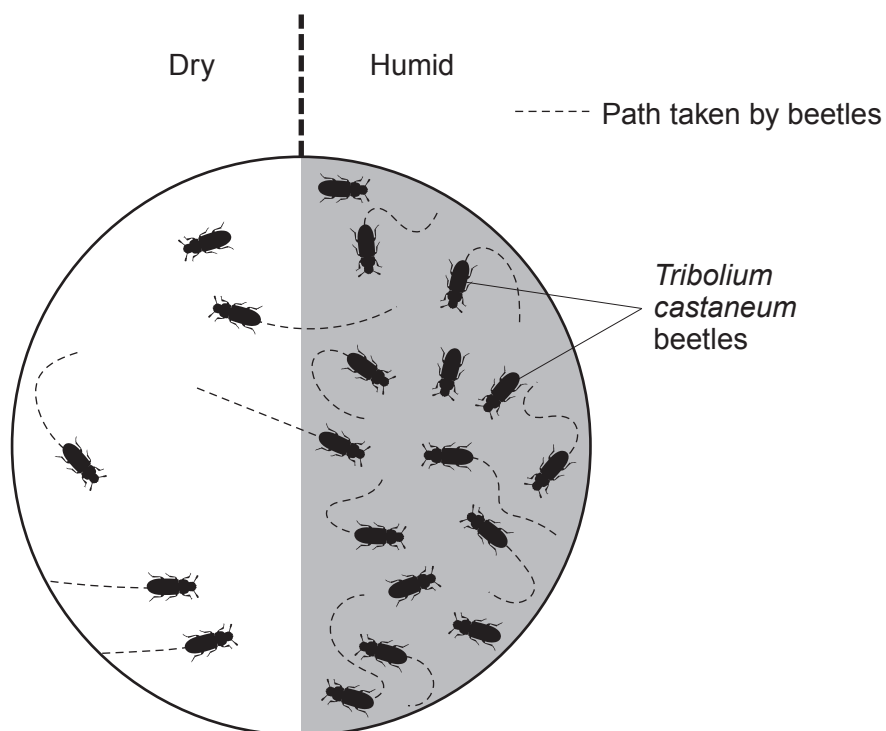
.....

.....

.....



- (iv) Students carried out the following experiment to investigate behaviour in insects. Twenty *Tribolium castaneum* beetles were placed in a Petri dish which had dry and humid areas. The number of beetles found in each area was recorded every 30 seconds for 3 minutes.



State the type of innate behaviour exhibited by the beetles in response to the environment they are placed in. [1]



- (v) The null hypothesis for this experiment was that there was no significant difference between the observed and expected number of *Tribolium* beetles found in the dry and humid areas. A Chi-squared test was then performed on the data to test the null hypothesis.

Degree of freedom	p = 0.1	p = 0.05	p = 0.02	p = 0.01
1	2.705	3.841	5.024	6.635
2	4.605	5.991	7.378	9.210
3	6.251	7.815	9.348	11.345
4	7.779	9.488	11.143	13.277
5	9.236	11.070	12.832	15.086

The calculated value of Chi-squared for these data was 5.00 with one degree of freedom. Use this information to reach a conclusion for this experiment at a suitable level of significance. [4]

.....

.....

.....

.....

.....

.....

END OF PAPER

20

